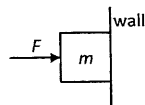


Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/6]

A block of mass m is held at rest by pushing it with a force F to prevent it from slipping.



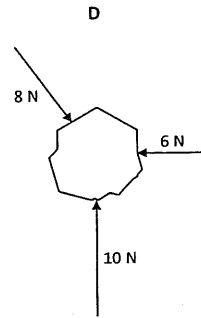
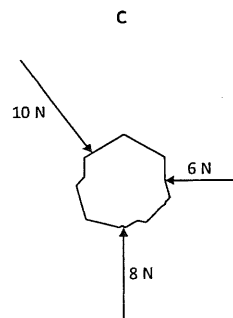
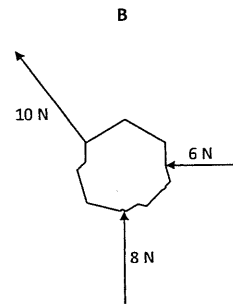
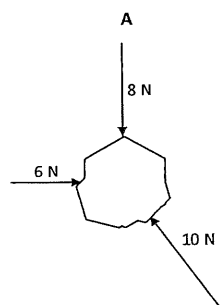
What is the magnitude of the contact force exerted by the wall on the block?

- A F B mg C $\sqrt{F^2 - m^2 g^2}$ D $\sqrt{F^2 + m^2 g^2}$

2. [ACJC/H2/Prelims 2018/P1/7 Modified]

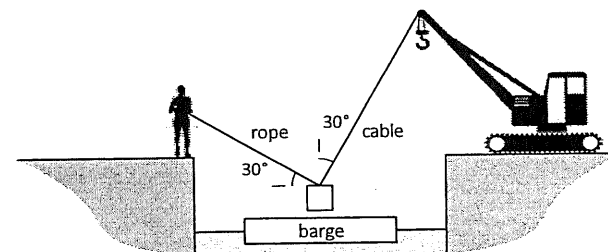
Three coplanar forces 10 N, 8 N and 6 N act on a flat object.

Which one of the following shows the flat object in equilibrium under the action of the forces?



3. [ACJC/H2/Prelims 2018/P1/8]

A mini crane is used to load a crate weighing 600 N onto a barge. The man uses a rope and applies a force of 300 N to the rope.

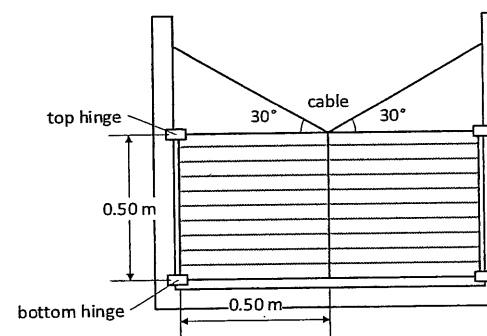


What is the tension in the cable from the crane?

- A 300 N B 520 N C 600 N D 690 N

4. [AIC/H2/Prelims 2018/P1/7]

A window is made up of 2 uniform panes. Each pane is 0.50 m wide and 0.50 m high, with hinges attached at the top and bottom as seen in the figure. A cable makes an angle of 30° with the top of the pane and has a tension of 150 N. The mass of one pane is 20 kg.

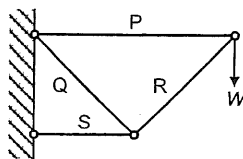


What is the magnitude and direction of the horizontal force exerted by the top hinge on the left pane?

- A 107 N to the right B 107 N to the left C 130 N to the right D 130 N to the left

5. [DHS/H2/Prelims 2018/P1/6]

In order to support a load W , four light hinged rods P, Q, R and S are connected as shown below and mounted in a vertical plane.

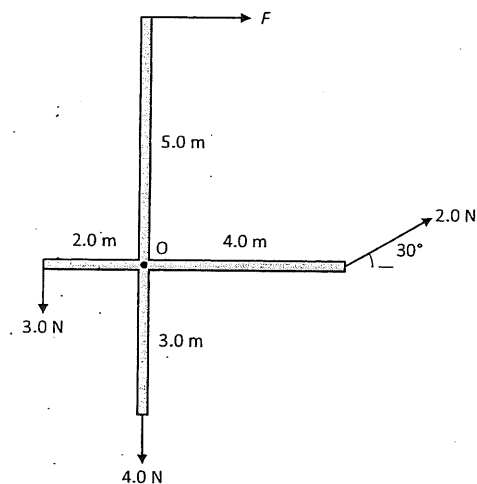


Which rods are in compression and which in tension?

	in compression	in tension
A	P	Q, R, S
B	P, Q	R, S
C	Q, R	P, S
C	R, S	P, Q

6. [JJC/H2/Prelims 2018/P1/6]

A cross-shaped structure, freely pivoted at O, has arms of lengths 5.0 m, 4.0 m, 3.0 m and 2.0 m. It is acted on by forces of 2.0 N, 3.0 N, 4.0 N and an unknown force F . The structure is in rotational equilibrium.

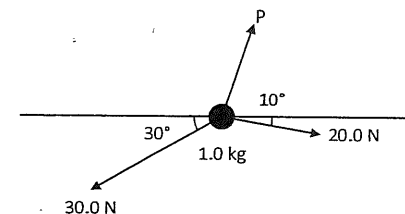


What is the magnitude of force F ?

- A 0.4 N B 2.0 N C 2.6 N D 4.4 N

7. [MI/H2/Prelims 2018/P1/7 Modified]

A 1.0 kg mass is at rest on a smooth horizontal surface. Three forces keep the 1.0 kg mass in equilibrium.

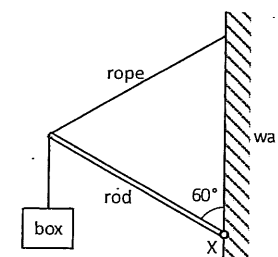


What is the magnitude of force P ?

- A 6.3 N B 12.2 N C 18.5 N D 19.5 N

8. [MI/H2/Prelims 2018/P1/9]

A uniform rod of weight 50 N and length L is hinged at X to a wall as shown. A rope of length L supports the rod and is also attached to the wall. A box of weight 24 N is suspended at the other end of the rod.

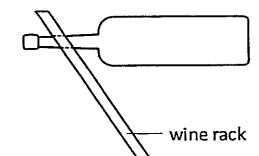


What is the tension in the rope?

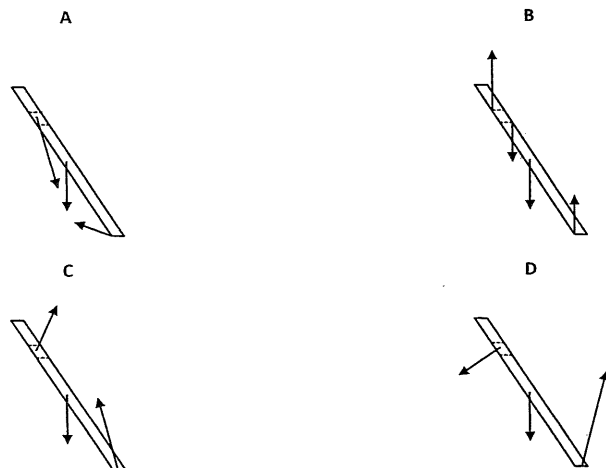
- A 28 N B 49 N C 74 N D 85 N

9. [NJC/H2/Prelims 2018/P1/5 Modified]

The diagram shows a wine rack with a bottle of wine that is balanced on the table.

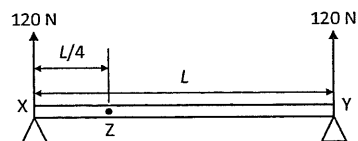


Which of the following diagrams correctly shows the directions of the forces acting on the wine rack?



10. [NYJC/H2/Prelims 2018/P1/7]

A uniform plank of length L is supported by initially equal forces of 120 N at X and Y.

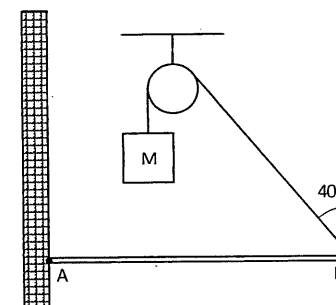


If the force at X is now moved to Z such that the plank is kept horizontal, what is the magnitude of the force at Y?

- A 60 N B 80 N C 120 N D 160 N

11. [PJC/H2/Prelims 2018/P1/4 Modified]

A thin uniform rod AB of mass 4.0 kg is suspended by a string attached at B. The string is connected to a block M and passed over a smooth pulley. The other end of the rod is hinged at A. When the rod is horizontal, the string makes an angle of 40° with the vertical.

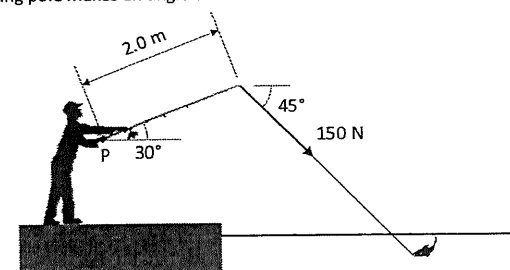


What is the weight of M required to maintain equilibrium?

- A 20 N B 26 N C 31 N D 39 N

12. [RVHS/H2/Prelims 2018/P1/5]

A man's 2.0 m long fishing pole makes an angle of 30° with the horizontal.

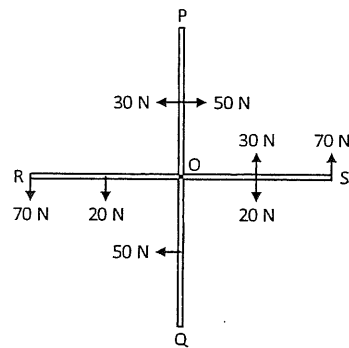


What is the moment exerted by the fish about the man's hand at P?

- A 75 N m B 77 N m C 110 N m D 290 N m

13. [SRJC/H2/Prelims 2018/P1/5]

A rigid cross-shaped structure having four arms PO, SO, QO and RO, each 1.80 m long, is pivoted at O. Forces act on the ends of the arms and on the midpoints of the arms as shown.



What is the net moment about O?

- A 54 N m B 108 N m C 216 N m D 432 N m

14. [SRJC/H2/Prelims 2018/P1/6]

Forces 5 N, 12 N and 13 N act at a point which is in equilibrium.

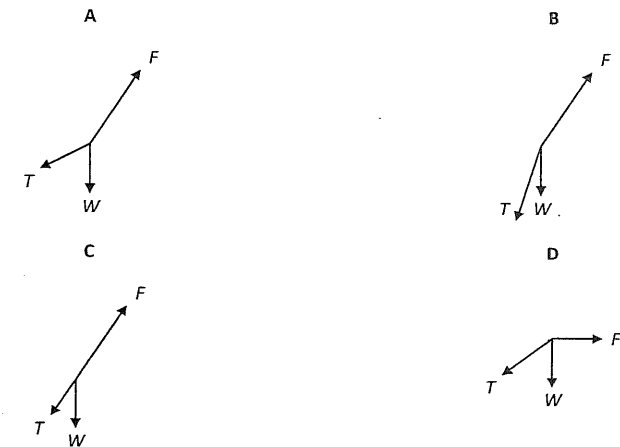
What is the angle between the 5 N and 13 N force?

- A 23° B 67° C 90° D 113°

15. [TPJC/H2/Prelims 2018/P1/5]

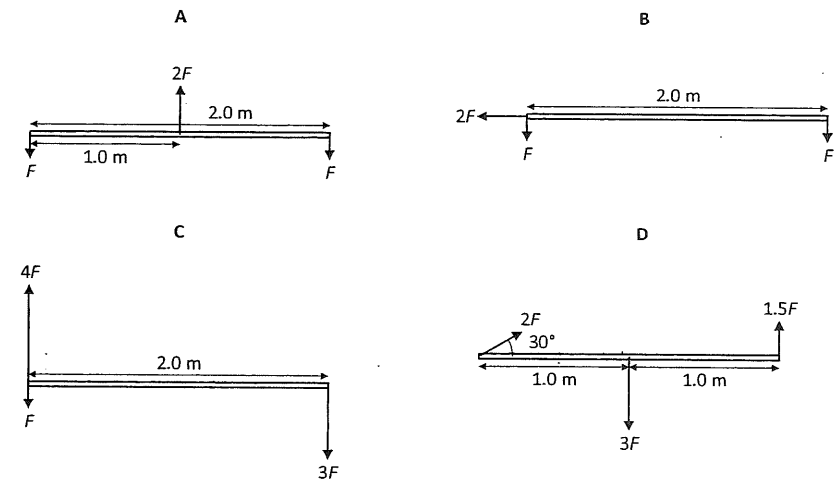
A kite is held stationary under the influence of three forces: the tension T in the string, the force F of the wind and the weight W of the kite.

Which one of the following force diagrams could be correct?



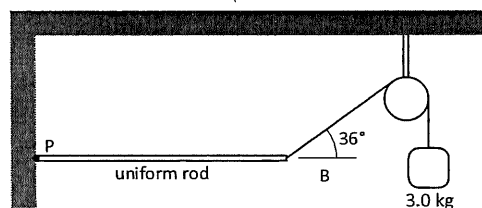
16. [TPJC/H2/Prelims 2018/P1/6]

In which situation will the forces applied to the rigid object **not** produce a resultant motion?



17. [TPJC/H2/Prelims 2018/P1/7 Modified]

The diagram below shows a uniform rod freely pivoted at P. The rod is suspended horizontally, when a string tied to one end of the rod passes over a smooth pulley and is attached to a 3.0 kg mass.

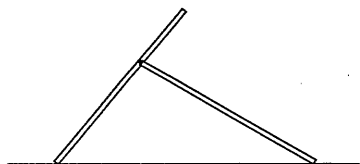


What is the mass of the rod?

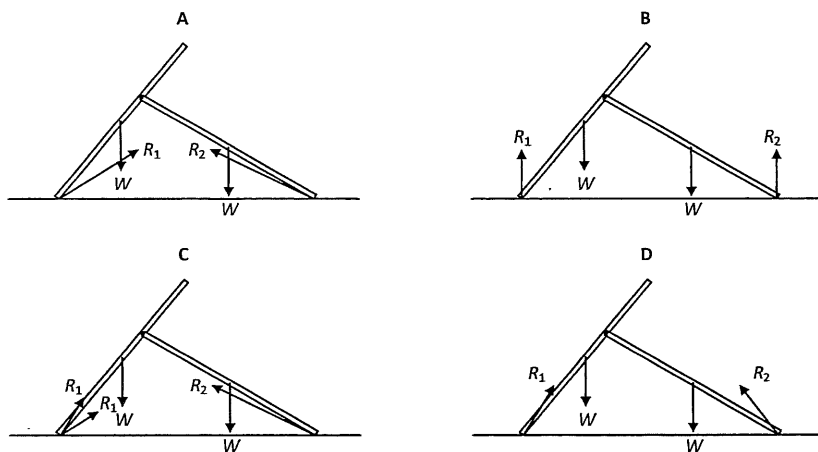
- A 4.8 kg B 3.5 kg C 2.4 kg D 1.7 kg

18. [VJC/H2/Prelims 2018/P1/3]

Two identical uniform rods, each of weight W , are hinged together to form a structure which is resting on a rough floor as shown.

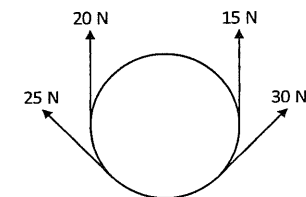


If the reaction forces acting on the structure by the floor are R_1 and R_2 , which of the following shows the forces acting on the structure?



19. [VJC/H2/Prelims 2018/P1/5]

Four coplanar forces act on a circular disc as shown.



What is the resultant force and resultant torque on the disc?

	net force	net torque
A	zero	non-zero
B	zero	zero
C	non-zero	non-zero
D	non-zero	zero

II. Structured Questions (ACJC – YJC)

1. [DHS/H2/Prelims 2018/P2/1 part]

- d. Two identical balls are placed in a smooth glass container as shown in Fig. 1.4. Each ball has a mass of 170 g. Their centres and point A lies on a straight line as shown by the dotted line.

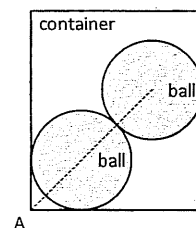


Fig. 1.4

- Determine the magnitude of the horizontal force by the container on the upper ball. [2]
- Hence, or otherwise, determine the magnitude of the force exerted by the lower ball on the upper ball. [2]

[Total: 12]

2. [HCI/H2/Prelims 2018/P3/1]

Fig. 2.1 is a diagram of a human arm lifting an object.

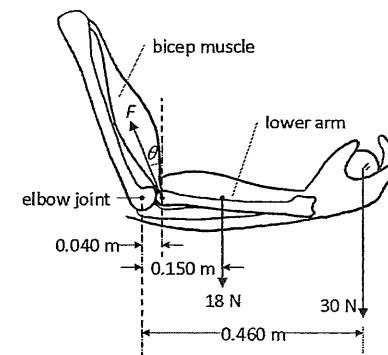


Fig. 2.1

The lower arm is horizontal and its centre of gravity is 0.150 m from the elbow joint. The weight of the lower arm is 18 N. The bicep muscle exerts a force F at an angle of θ to the vertical.

The horizontal distance between the elbow joint and the point of attachment of the muscle to the lower arm bone is 0.040 m. The weight of the object held in the hand is 30 N and its centre of gravity is 0.460 m from the elbow joint. The arm is in equilibrium.

- Define *centre of gravity*. [1]
- Determine the value of F when $\theta = 15^\circ$. [2]
- For the lower arm to be in equilibrium, the elbow joint also needs to exert a force R on the lower arm bone.
 - Draw a labelled arrow on Fig. 2.1 to represent the force R that the elbow exerts on the lower arm. [1]
 - Explain the direction of this force R . [2]
- As the lower arm is slowly moved away from the body in the horizontal direction, the angle θ increases.
 - Sketch in Fig. 2.2 the graph of the variation with θ of F .
You may assume that the lower arm remains horizontal and is in equilibrium at all times.

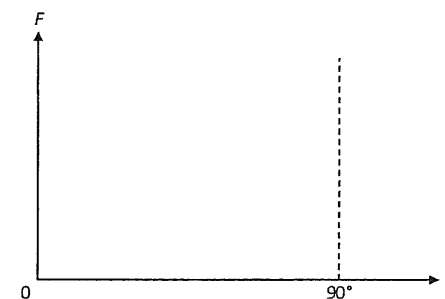


Fig. 2.2

- Explain the shape of your graph in Fig. 2.2.

[2]
[2]
[Total: 10]

3. [JJC/H2/Prelims 2018/P2/2 Modified]

- a. State the conditions for a body to be in equilibrium. [2]
- b. Fig. 3.1 shows a 70 kg climber hanging by only the crimp hold of one hand on the edge of a shallow horizontal ledge in a rock wall. The fingers are pressed down. Her feet touch the rough rock wall at distance $H = 2.0$ m directly below her crimped fingers. The frictional force between her feet and the wall is 32 N. Her centre of mass is at distance $a = 0.20$ m from the wall.

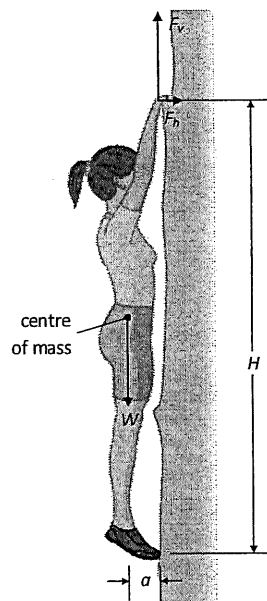


Fig. 3.1

- i. Determine the horizontal frictional force F_h acting on the fingers. [2]
- ii. Determine the vertical force F_v acting on the fingers. [1]
- iii. The weight W of the climber and the resultant force F exerted by the ledge on her fingers are indicated in Fig. 3.1. Draw and label the force S exerted by the rock wall on her feet. [2]

[Total: 7]

4. [MI/H2/Prelims 2018/P3/1]

- a. Fig. 4.1 shows the variation of force F with the extension x of a spring.

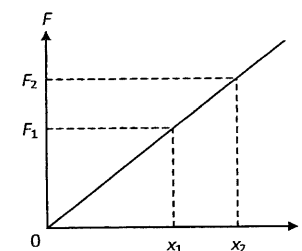


Fig. 4.1

The extension of the spring is increased from x_1 to x_2 .

Show that the work done W in extending the spring is given by

$$W = \frac{1}{2}k(x_2^2 - x_1^2)$$

where k is the spring constant.

[2]

- b. Fig. 4.2 shows a signboard suspended by two elastic springs.

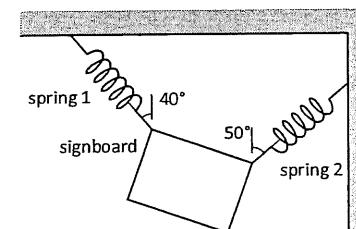


Fig. 4.2

- i. State the conditions for equilibrium. [2]
- ii. On Fig. 4.2, draw and label clearly the forces acting on the signboard. Mark the centre of gravity of the signboard with a dot and label the point as G. [2]
- iii. Given that the tension in spring 1 is 300 N and the tension in spring 2 is 252 N, determine the weight of the signboard. [2]
- iv. The signboard is pulled vertically downwards with a force of 20 N so that the springs are stretched to a new position. Determine the acceleration of the signboard immediately after it is released. [2]

[Total: 10]

5. [MJC/H2/Prelims 2018/P3/1]

- a. i. State Newton's first law of motion. [1]
- ii. State the conditions for equilibrium. [2]
- b. Fig. 5.1 shows a uniform ladder of weight 80 N resting on a smooth wall and a rough floor. The ladder makes an angle of 60° with the floor.

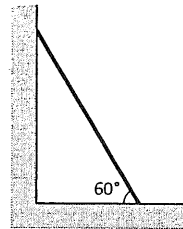


Fig. 5.1

- i. Show that the force exerted by the wall on the ladder is 23 N. [2]
- ii. Calculate the force exerted by the floor on the ladder. [3]
- iii. A person now stands on the ladder. The ladder remains stationary. State and explain the effects, if any, on
 1. the vertical force exerted by the floor on the ladder, [1]
 2. the horizontal force exerted by the wall on the ladder. [1]

[Total: 10]

6. [NYJC/H2/Prelims 2018/P3/2]

A person supports a load of 20 N in his hand as shown in Fig. 6.1. The system consisting of the arm and load is equivalent to the system shown in Fig. 6.2. The rod represents the forearm, and T represents the tension exerted in the biceps. The forearm weighs 65 N.

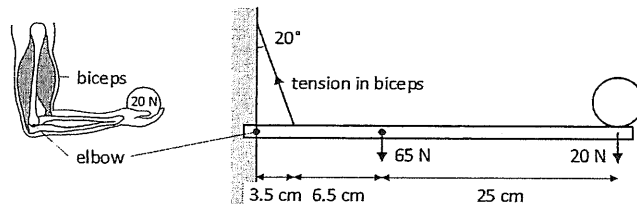


Fig. 6.1

Fig. 6.2

- a. Show that the tension T in the biceps is 410 N. [2]
- b. Hence, determine the magnitude and direction of the force acting at the elbow. [4]

[Total: 6]

7. [SRJC/H2/Prelims 2018/P2/2]

A uniform rectangular card is suspended from a wooden rod. The card is held at one of its ends as shown in Fig. 7.1. The force by the hand on the card acts horizontally to the right.

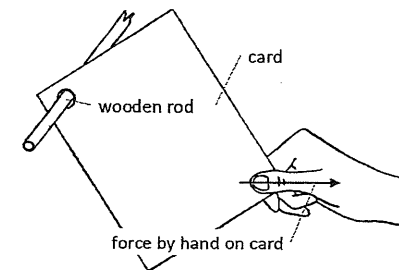


Fig. 7.1

- a. On Fig. 7.1,
 - i. mark with an 'X' the position of the centre of gravity of the card. [1]
 - ii. draw an arrow labelled with W to represent the weight of the card. [1]
- b. State the conditions for the card to be in equilibrium. [2]
- c. Draw an arrow labelled with R on Fig. 7.1 to represent the force exerted by the wooden rod on the card. Show your construction clearly. [2]
- d. The card is now released. It swings on the wooden rod and eventually comes to a rest. By reference to the completed diagram in Fig. 7.1, describe the final position in which the card comes to a rest. [1]

[Total: 7]

8. [TPJC/H2/Prelims 2018/P2/1 Part]

- a. A uniform rigid rod of weight 400 N is attached to a vertical beam by a hinge as shown in Fig. 8.1. The other end of the rod is fastened to a support cable. The structure is used to support a load of weight 2000 N.

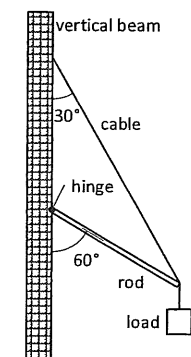


Fig. 8.1

- i. State two conditions needed for the rod to be in a state of equilibrium. [2]
- ii. Determine the magnitude of the tension in the support cable. [3]

[Total: 10]

9. [VIC/H2/Prelims 2018/P2/1]

- b. A hungry bear walks onto a horizontal uniform beam of length L , supported on the left by a hinge attached to a wall, in an attempt to retrieve some “goodies” hanging at the right end (Fig. 9.2). The total weight of the bear and the beam is 900 N. The goodies weigh 80.0 N. Fig. 9.3 gives the tension T in the cable as a function of the bear’s position given as a fraction x/L of the beam length.

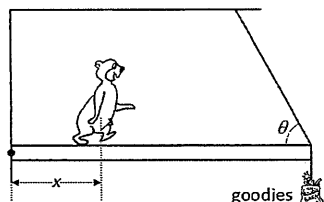


Fig. 9.2

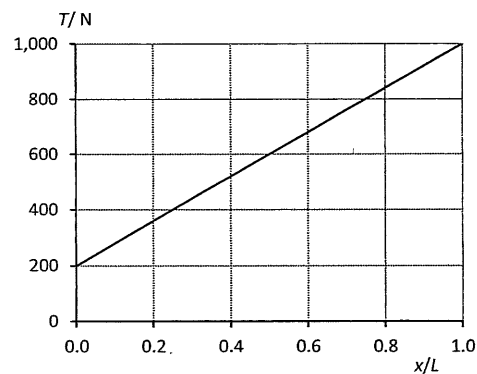


Fig. 9.3

- By considering the bear standing at the centre of the beam, calculate the angle θ . [2]
- Suggest where, on the beam, the bear should stand so that his weight does not produce a moment on the beam. [1]
- Calculate the mass of the beam. [2]

[Total: 5]

10. [YJC/H2/Prelims 2018/P2/2]

- Explain what is meant by the following terms when used in the context of forces.
 - Centre of gravity. [1]
 - Friction. [1]
- Fig. 10.1 shows a man performing a bungee jump.

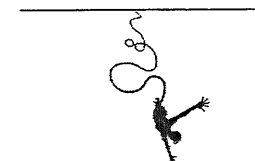


Fig. 10.1

He falls 42 m before the elastic rope secured to him starts to exert some force on him. The force-extension graph for this rope is shown in Fig. 10.2.

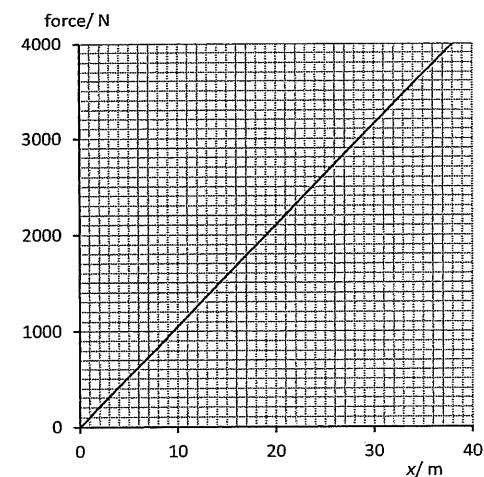


Fig. 10.2

The total distance the man falls before he stops for the first time is 78 m.

- Deduce
 - the extension of the rope when the man stops for the first time, and [1]
 - the elastic potential energy stored in the rope at this time. [2]
- Show that the man has a mass of approximately 90 kg. [2]

[Total: 7]

